

HOME AUTOMATION COOKBOOK

Starter House Setup Guide

One hub. About ten devices. One working recipe per room. A printable plan for your first month.

homeautomationcookbook.com

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How to use this guide

Print it. Keep it near the hub. Check boxes as you go.

Do one room at a time. Finish a room, live with it for a week, then move on. A slow start beats a house nobody trusts.

Every walkthrough is written in plain steps that work on any hub. Where a step is specific to Hubitat, it says so. Each one links to the full recipe on the website for platform notes and failure modes.

Amazon product links in this guide are affiliate links, and the site earns from qualifying purchases. See homeautomationcookbook.com/disclosure.html. Check that every device works with your hub before you buy. Follow the maker's instructions and local electrical codes. Hire an electrician for wiring you are not comfortable with.

The rules for a starter house

> **One hub.** Do not mix three apps on day one.

> **Local first.** Prefer Z-Wave and Zigbee devices that keep working when the internet is down.

> **Switches still work.** Every light keeps a normal wall control.

> **One room at a time.** Finish a room, live with it for a week, then move on.

> **No risky jobs yet.** Leave locks, heaters, and garage door control for later.

Pick the hub

Hubitat runs automations on the hub itself. It supports Z-Wave and Zigbee. It lets you build simple rules now and detailed ones later. This is the hub the site runs on.

SmartThings is a reasonable alternative, especially if you already own Samsung devices.

Whichever you pick, do these three things on day one:

- [ ] Give the hub a fixed IP address on your router.
- [ ] Turn on its backups.
- [ ] Install the hub's app on at least two phones, so alerts reach more than one person.

Read more: homeautomationcookbook.com/getting-started/choose-the-right-platform.html

Naming devices

Name every device by room and job. "Laundry Washer Power" is better than "ZEN15". Good names make every rule easier to read, and every alert easier to act on.

Mixing Zigbee and Z-Wave is fine. Mains-powered switches and plugs act as repeaters and make the mesh stronger. Install those before the battery sensors in far rooms.

The four-weekend plan

When	Room	Devices	Recipes
Weekend 1	Hallway or laundry lights	Switch, motion sensor	Lights on motion. A week later, lights off after motion.
Weekend 2	Laundry	ZEN15, vibration sensor	Washer done, dryer done. Log three loads first.
Weekend 3	Bathroom	Multisensor, dimmer	Bathroom night-light
Weekend 4	Safety and status	Leak sensors, contact sensor, tilt sensor	Leak alerts, closet light, garage open alert, low-battery alerts
After that	Whole house	None new	Modes, away mode, bedtime routine

Shopping list

These are the products the website recommends for a starter house. Pick the "good enough" option or the step up from the device guide at homeautomationcookbook.com/getting-started/device-guide.html.

	Job	Product	Where it goes	Qty
☐	Hub	Hubitat or SmartThings (see "Pick the hub")	Central spot, wired to the router	1
☐	Motion sensor	Aeotec SmartThings Motion Sensor	Hallway or laundry	1
☐	Smart switch	Leviton DG15S Zigbee switch	Same room as the motion sensor	1
☐	Bathroom multisensor	Aeotec TriSensor 8	Bathroom	1
☐	Smart dimmer	TP-Link Tapo S505D	Bathroom or living room	1
☐	Power-metering switch	Zooz ZEN15	Washer	1
☐	Vibration sensor	THIRDREALITY Zigbee Vibration Sensor	Dryer	1
☐	Leak sensor	THIRDREALITY WL2	Water heater, washer, sink	2
☐	Contact sensor	SONOFF SenseGuard Gen2	Closet or pantry door	1
☐	Garage tilt sensor	THIRDREALITY Garage Tilt Sensor	Garage door	1

That is ten devices plus the hub. Prices and stock change, so check the current listing. Also buy: spare batteries for each sensor, and double-sided mounting tape.

Before you order

- ☐ Each device is listed as working with your hub.
- ☐ Switch and dimmer boxes have the wiring the device needs (check for a neutral wire if required).
- ☐ The washer's load is within the ZEN15 rating.
- ☐ Buy one of each first. Buy more of what works.

Room-by-room checklists

Whole house

- [ ] Hub set up and updated
- [ ] Fixed IP address set on the router
- [ ] Hub backups turned on
- [ ] Modes created: Day, Evening, Night, Away
- [ ] Notification app installed for at least two people
- [ ] Mains switches and plugs paired before far battery sensors
- [ ] Every device named by room and job

Hallway or laundry lights

- [ ] Smart switch installed, paddle works without the hub
- [ ] Motion sensor placed to see the entry path
- [ ] Lights on motion built and tested at night
- [ ] Lights on motion tested in daylight (stays off)
- [ ] Lived with it for one week
- [ ] Lights off after motion added
- [ ] Sat still past the timeout to test

Laundry

- [ ] ZEN15 on the washer, load within its rating
- [ ] Vibration sensor on the dryer
- [ ] Three loads logged before alerts
- [ ] Thresholds written down
- [ ] Washer alert gives one notification per load
- [ ] Dryer alert gives one notification per load

Bathroom

- [ ] Dimmer installed, paddle still works
- [ ] Multisensor away from the shower plume and towels
- [ ] Night-light at 10 percent in Night mode
- [ ] Off timeout tested during a shower
- [ ] Optional: humidity reading checked for a future fan recipe

Kitchen and water

- [ ] Leak sensor under the water heater
- [ ] Leak sensor behind the washer or dishwasher
- [ ] Optional: leak sensor under the kitchen sink
- [ ] Wet towel test done for every sensor
- [ ] Two people received the alert

Closet, pantry, and garage

- [ ] Contact sensor parts lined up
- [ ] Door tested ten times in a row
- [ ] Tilt sensor on the top garage panel
- [ ] 10 minute open alert tested
- [ ] Bedtime garage check tested

Before you call it done

- [ ] Low-battery alert tested with a weak battery
- [ ] Away mode tested with everyone gone and with one person home
- [ ] Night mode tested by hand and by the backup rule
- [ ] Someone else in the house can work the lights without the app
- [ ] Device names and thresholds written on the notes page

Step-by-step rule walkthroughs

Each walkthrough uses the same shape: a trigger, conditions, and actions. Build the smallest version first. Add extras only when daily use shows you need them.

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1. Lights on motion

The light comes on before your hand reaches the switch. The switch still works.

1. Install the switch first and check that the paddle works like a normal switch.
2. Pair the motion sensor. Name it by room and job, for example "Hallway Motion".
3. Mount the sensor where it sees the path into the room, not the wall switch.
4. Create the rule. Trigger: motion becomes active.
5. Add a condition: the light is off. This keeps the rule from changing a level someone picked.
6. Add a second condition: the room is dark, or it is inside your night hours.
7. Action: turn on the light. At night use 10 to 20 percent. In the evening use a useful level, such as 60 percent.
8. Do not add a turn-off step yet. That is the next walkthrough.

> **Test it:** Walk in from the dark at night. The light should be on before you reach the switch. Then walk in on a bright day. It should stay off.

Tip: On Hubitat, Room Lighting is the natural starting point. A Basic Rule also works for a simple on action.

Full recipe: www.homeautomationcookbook.com/automation/lighting/lights-on-motion.html

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2. Lights off after motion

The light turns off after the room has been empty for a while, but never while someone is still there.

1. Live with recipe 1 for a week before you add this one.
2. Trigger: the room sensor has reported no motion for the timeout.
3. Start the timeout at 5 to 10 minutes for a bathroom or quiet room. Use 15 minutes if people sit still.
4. Condition: no other sensor in the same space is active. If a closet opens into the room, watch both sensors.
5. Condition: no manual override. If someone used the wall switch, leave the light alone.
6. Optional condition: music is not playing in the room. Playing music is a good sign someone is still there.
7. Action: turn the light off.
8. Make sure new motion cancels the countdown.

> **Test it:** Sit still in the room for longer than the timeout. If the light goes off on you, raise the timeout before you blame the sensor.

Tip: Shorten the timeout only after the household has lived with it for a while.

Full recipe: www.homeautomationcookbook.com/automation/lighting/lights-off-after-motion.html

3. Bathroom night-light

A dim, warm light at night so nobody is blinded at 2 AM.

1. Install the dimmer and pair the multisensor.
2. Trigger: bathroom or entry motion is detected.
3. Condition: the mode is Night, or the time is inside your night window.
4. Optional condition: the room is dark enough, using the sensor's light reading.
5. Action: set the bathroom light to 10 percent. Add a warm color only if the light supports it.
6. Let recipe 2 own the turn-off. Start at 10 minutes for a shower bathroom.

> **Test it:** Walk in at night with the house in Night mode. The light should come on dim. A press on the switch should still bring full light.

Tip: If you later add the shower fan recipe, the same multisensor gives you the humidity reading.

Full recipe: www.homeautomationcookbook.com/automation/lighting/bathroom-night-light.html

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4. Washer done alert

One alert when the washer is really done. No alerts during pauses in the cycle.

1. Plug the washer into the ZEN15. Check that the washer is within the switch's rated load.
2. Run three loads with logging only. Write down the power while running, during pauses, and when finished.
3. Pick a running threshold that is clearly above idle power. Pick a finished threshold just above the idle level.
4. Rule A. Trigger: power stays above the running threshold long enough to prove a cycle started. Action: set a "Washer running" marker.
5. Rule B. Trigger: power stays below the finished threshold longer than the longest normal pause.
6. Rule B condition: the "Washer running" marker is set.
7. Rule B actions: send one "Washer finished" notification, set "Laundry waiting", and clear the running marker.
8. Never let the rule turn the plug off.

> **Test it:** Run a normal load and watch your phone. You should get exactly one alert, a few minutes after the washer stops.

Tip: On Hubitat, the free Appliance Cycle Monitor app handles the running and finished states for you.

Full recipe: www.homeautomationcookbook.com/automation/appliances/washer-done-notification.html

5. Dryer done alert

One alert when the dryer stops shaking for good.

1. Stick the vibration sensor on top of the dryer, away from the hot vent.
2. Run a load and note how long vibration stops during normal pauses or cooldown.
3. Rule A. Trigger: vibration stays active for the start delay. Action: set a "Dryer running" marker.
4. Rule B. Trigger: vibration stays inactive longer than the longest pause you saw.
5. Rule B conditions: the "Dryer running" marker is set, and the sensor is online.
6. Rule B actions: send one "Dryer finished" notification, set "Laundry waiting", and clear the marker.

> **Test it:** Run a full load, including any wrinkle guard. You should get one alert, not three.

Tip: If you get alerts during cooldown, the finish delay is too short.

Full recipe: www.homeautomationcookbook.com/automation/appliances/dryer-done-notification.html

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6. Water leak alert

An alert at any hour that says exactly where the water is.

1. Name each sensor by place, for example "Leak Water Heater" and "Leak Washer".
2. Place them flat on the floor where water shows up first.
3. Trigger: any leak sensor reports wet.
4. No conditions. Leak alerts run in Home, Away, Night, and Guest modes.
5. Action: send a high-priority notification that names the sensor. Send it to at least two people.
6. Optional: repeat the alert only while nobody has checked on it.
7. Clear the incident by hand, after a person has looked at the source. A dry sensor does not mean the leak is fixed.

> **Test it:** Touch the sensor with a wet towel. Check that both phones get the alert and that the message names the right place.

Tip: Leave automatic water shutoff for later. Get trusted alerts first.

Full recipe: www.homeautomationcookbook.com/automation/security/water-leak-response.html

7. Closet light on door open

The closet or pantry light comes on when the door opens.

1. Mount the contact sensor on the door and frame with the two parts lined up.
2. Trigger: the contact changes from closed to open.
3. Condition: the space is dark enough, or it is inside the hours when light makes sense.
4. Action: turn on the light at the level you want.
5. Optional second rule: turn the light off when the door closes, or after a timeout.
6. Do not undo a manual switch change right away.

> **Test it:** Open the door ten times in a row. The light should come on every time.

Tip: If it misses, check the gap between the two sensor parts.

Full recipe: www.homeautomationcookbook.com/automation/lighting/toggle-lights-door.html

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8. Garage door left open

A reminder when the garage door has been open longer than expected.

1. Mount the tilt sensor on the top panel of the garage door.
2. Trigger: the door reports open and stays open for 10 minutes.
3. Action: send "Garage door has been open for 10 minutes".
4. Second rule. Trigger: 10:00 PM. Condition: the door is open. Action: send a high-priority "Garage door still open at bedtime".
5. Do not close the door from this recipe. Closing stays manual.

> **Test it:** Open the garage door and wait 10 minutes. Then close it and make sure no more alerts arrive.

Tip: Auto-closing is a separate, later project. It needs an opener that supports safe unattended operation.

Full recipe: www.homeautomationcookbook.com/automation/security/garage-door-notification.html

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9. Low-battery alerts

No sensor ever goes quiet without anyone knowing.

1. Add every battery device to one list.
2. Trigger: a battery drops below its threshold, or reports low battery.
3. Condition: this device has not been flagged in the last 7 days.
4. Action: send "[Device name] battery is at [level]".
5. The alert resets when the battery is replaced and reports normal again.

> **Test it:** Put a weak battery in one sensor. You should get one alert, not one every hour.

Tip: Keep spare batteries of each size in one drawer.

Full recipe: www.homeautomationcookbook.com/automation/notifications/low-battery-alerts.html

10. Away mode

The house switches to Away only when it is really empty.

1. Set up presence for each person and test it for a week.
2. Trigger: the last tracked person leaves.
3. Condition: everyone has been away for 15 minutes.
4. Condition: Guest mode is off, and no recent indoor motion suggests someone is home.
5. Actions: set the mode to Away, turn off lights that are safe to turn off, and send a summary.
6. Cancel it if someone returns, or if a person picks Home or Guest.

> **Test it:** Leave together and check that Away is set after the delay. Then leave one person home and check that it is not.

Tip: Shorten the 15 minutes only after several weeks without a false Away.

Full recipe: www.homeautomationcookbook.com/automation/daily-routines/away-mode.html

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11. Bedtime and Night mode

Night mode from one intentional action, with an automatic backup.

1. Manual rule. Trigger: a double tap down on a stair or hallway switch.
2. Conditions: the mode is Home, and the time is between 8 PM and midnight.
3. Action: set the mode to Night and turn off approved unoccupied lights. Keep path lighting.
4. Automatic backup. At 10 PM, if the mode is Home, check every 10 minutes, up to 12 times.
5. Each pass: stop if the mode is no longer Home. If every downstairs motion sensor has been inactive for 15 minutes, set Night and stop.
6. If people are still up after about two hours, stay in Home until someone sets Night by hand.
7. Report anything that still needs a person, like an open garage door.

> **Test it:** Double tap the switch and check that the mode changes. On another night, go to bed without it and check that the backup sets Night.

Tip: The double tap is the main action. The automatic rule only catches the nights when everyone forgets.

Full recipe: www.homeautomationcookbook.com/automation/daily-routines/bedtime-routine.html

Add modes, then routines

Once the devices work on their own, add home modes. Day, Evening, Night, and Away are enough.

Modes make every later recipe simpler. A rule can check one mode instead of five conditions.

Build Away mode and the bedtime routine (walkthroughs 10 and 11) only after the room recipes have run well for a few weeks.

What to skip in the first month

> **Smart locks and auto-closing garage doors.** Get alerts working first.

> **Plugs on heaters or cooking appliances.** Not safe for general-purpose plugs.

> **Voice-only control.** Fine as an extra, not as the only way to turn on a light.

> **Bundles of cheap devices.** Test one, then buy more of what works.

Troubleshooting

Problem	First thing to try
Light turns on after you reach the switch	Move the sensor so it sees the entry path earlier.
Light turns on in daylight	Add a light-level condition. Test on a bright overcast day too.
Light turns off while someone sits still	Raise the timeout, or watch a second sensor.
Washer or dryer alert comes more than once	The finish delay is shorter than a normal pause. Make it longer.
No washer alert at all	Check that the "running" marker is set during a load.
A battery sensor drops off	Add a mains-powered switch or plug between it and the hub.
A rule stops working when the internet is down	Check that the devices and rule run locally on the hub.

Notes

Device name	Room	Threshold or timeout	Battery size

Hub details

Hub IP address:

Last backup date:

Washer running / finished watts:

Dryer longest pause:

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